

Active Connections Analysis Using Netstat

Objective

The purpose of this study is to analyze active network connections, listening ports, and connection states using the netstat command. The objective is to gain practical experience in monitoring network activity and interpreting TCP/IP connection information for network monitoring and basic network troubleshooting.

Tools Used

Commands used in this study:

- netstat
- tasklist

Step 1 – Active Connection Analysis

The command was executed: netstat -an

Output (Representative Excerpt):

The complete `netstat -an` output contained numerous active connections and listening ports. The excerpt below highlights representative examples of the most relevant connection states observed during the analysis.

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING
TCP	0.0.0.0:5432	0.0.0.0:0	LISTENING
TCP	:::135	:::0	LISTENING
....			
TCP	192.168.0.x:49419	98.xx.xx.xx:443	ESTABLISHED
TCP	192.168.0.x:49891	162.xx.xx.xx:443	ESTABLISHED
....			
TCP	192.168.0.x:49671	46.xx.xx.xx:53	TIME_WAIT
TCP	192.168.0.x:52664	40.xx.xx.xx:443	TIME_WAIT
....			

```

TCP    192.168.0.x:52504    128.xx.xx.xx:443    CLOSE_WAIT
TCP    192.168.0.x:63412    128.xx.xx.xx:443    CLOSE_WAIT
....
UDP    0.0.0.0:5353         *.*
UDP    192.168.0.8:137      *.*
UDP    [::]:123             *.*

```

Findings:

The `netstat -an` output was examined to identify active network connections, listening ports, and TCP connection states.

- TCP and UDP protocols were observed.
- Multiple services were found in the **LISTENING** state.
- Active **ESTABLISHED** connections to external network services were identified.
- Several **TIME_WAIT** connections were observed, indicating recently closed TCP sessions.
- A small number of **CLOSE_WAIT** connections were present.
- Local loopback (`127.0.0.1`) connections were detected, indicating inter-process communication on the same computer.

Analysis:

The `netstat -an` output shows that the system maintains multiple active TCP connections while several services are listening for incoming requests. Active TCP connections, including secure HTTPS connections on port 443, indicate ongoing communication with external network services. The presence of `TIME_WAIT` states demonstrates the normal TCP connection termination process, while loopback connections (`127.0.0.1`) show communication between applications running on the same computer. During the analysis, both IPv4 and IPv6 entries were observed. IPv4 addresses appeared in standard dotted-decimal notation, while IPv6 addresses were displayed in square brackets. This indicates that the operating system supports dual-stack networking, allowing communication over both IP versions. Overall, the results provide valuable insight into active network sessions, listening services, and TCP connection states, demonstrating how the `netstat` command can be used for network monitoring and basic troubleshooting.

Step 2 – Process Identification

The command was executed: `netstat -ano`

Output (Representative Excerpt):

Active Connections

Proto	Local Address	Foreign Address	State	PID
TCP	0.0.0.0:5432	0.0.0.0:0	LISTENING	5436

TCP	192.168.0.x:49422	98.xx.xx.xx:443	ESTABLISHED	5332
TCP	192.168.0.x:52017	3.xx.xx.xx:443	TIME_WAIT	0
TCP	:::135	:::0	LISTENING	2012
UDP	0.0.0.0:123	*.*		28956

Findings:

The `netstat -ano` output was examined to associate active network connections with their corresponding Process IDs (PIDs).

- Each active connection was associated with a Process ID (PID).
- Both IPv4 and IPv6 connections included PID information.
- Listening services and established connections could be mapped to running processes.
- UDP entries also displayed the owning process ID.

Analysis:

The additional `PID` column provides a direct link between network connections and the processes that own them. This information is particularly useful during network troubleshooting and security investigations because it allows administrators to identify which application is responsible for a specific connection or listening port. By combining `netstat -ano` with `tasklist` or Windows Task Manager, a process can be identified and investigated further if necessary.

*** The identified PIDs were then verified using the `tasklist` command to determine the corresponding running applications.**

Step 3 – Process Verification

The command was executed: `tasklist /FI "PID eq <PID>"`

Output:

PID	Process	Description
5436	postgres.exe	PostgreSQL database service
5332	svchost.exe	Windows Service Host process
21584	steam.exe	Steam client application
4	System	Windows kernel/system process

Findings:

The `tasklist` command was used to identify the applications associated with selected Process IDs (PIDs) obtained from the `netstat -ano` output.

- PID **5436** was identified as **PostgreSQL (postgres.exe)**, confirming that the database service was listening on its default port (**5432**).
- PID **5332** corresponded to **svchost.exe**, a Windows Service Host process responsible for running one or more system services.
- PID **21584** belonged to **Steam (steam.exe)**, confirming that the observed network connections were created by the Steam client.
- PID **4** was identified as **System**, indicating that the listening port was owned by the Windows operating system.

Analysis:

The `tasklist` command successfully mapped Process IDs (PIDs) to their corresponding applications. This confirmed that different types of network connections were generated by different processes, including operating system services (System and `svchost.exe`), a database service (`postgres.exe`), and a user application (`steam.exe`). Combining `netstat -ano` with `tasklist` enables administrators to identify which application is responsible for a specific network connection or listening port, making these commands valuable tools for network monitoring, troubleshooting, and security analysis.

Key Concepts Learned

During this exercise, the following concepts were reinforced:

- TCP vs UDP
- Local Address
- Foreign Address
- Port Number
- Connection states (LISTENING, ESTABLISHED, TIME_WAIT, CLOSE_WAIT)
- Loopback Address (127.0.0.1)
- IPv4 and IPv6 Addressing
- Ephemeral Ports
- Active Network Connections
- Process ID (PID)
- Process-to-Port Mapping
- Network Troubleshooting with `netstat` and `tasklist`

Conclusion

This lab demonstrated how `netstat` can be used to monitor active network connections, identify listening ports, and examine TCP connection states. By using the `-ano` option together with `tasklist`, it was possible to associate network connections with the processes responsible for them. The exercise also reinforced the understanding of IPv4 and IPv6 addressing, loopback communication, and the role of PIDs in network analysis. Overall, these commands provide a practical foundation for network monitoring, troubleshooting, and basic security investigations in Windows environments.